

Comparing Machine Learning Classifiers for Breast Cancer Diagnosis

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Introduction and dataset

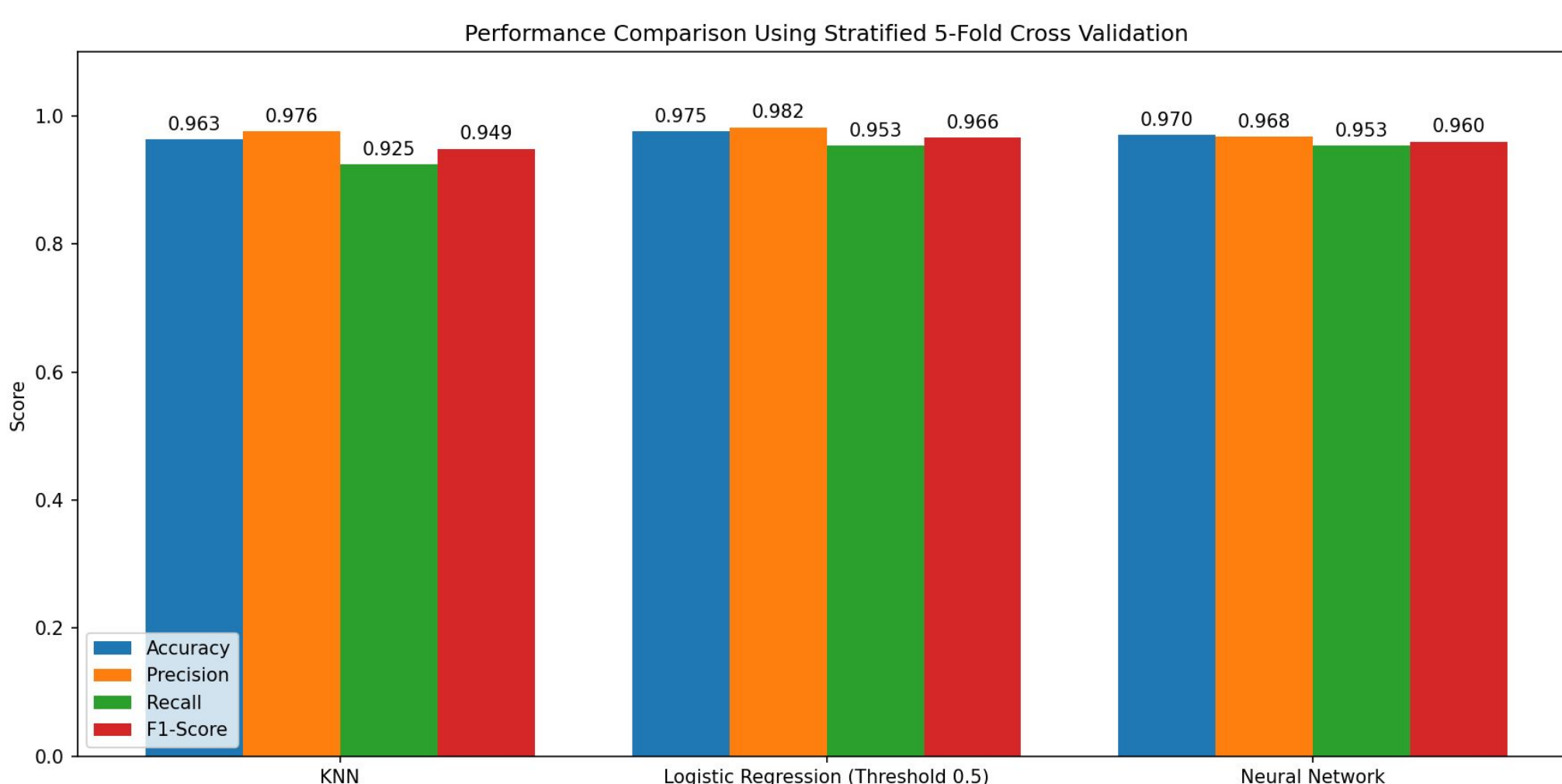
- Breast cancer detection is critical for patient survival
- Missing malignant cases (FN) is very dangerous and machine learning can assist in early diagnosis
- **Goal:** Compare ML models and evaluate decision thresholds
- **Dataset:** Wisconsin Breast Cancer Dataset
- 569 samples, 30 features
- Labels: 1 - Malignant, 0 - Benign

Method

- **Models:**
 - K-Nearest Neighbors (KNN)
 - Logistic Regression
 - Neural Network (PyTorch)
- **Evaluation:**
 - Stratified 5-fold cross-validation
- **Metrics:**
 - Accuracy, Precision, Recall, F1-score

Result

- Logistic regression achieved best overall performance
- Neural network performed competitively
- KNN showed slightly lower performance

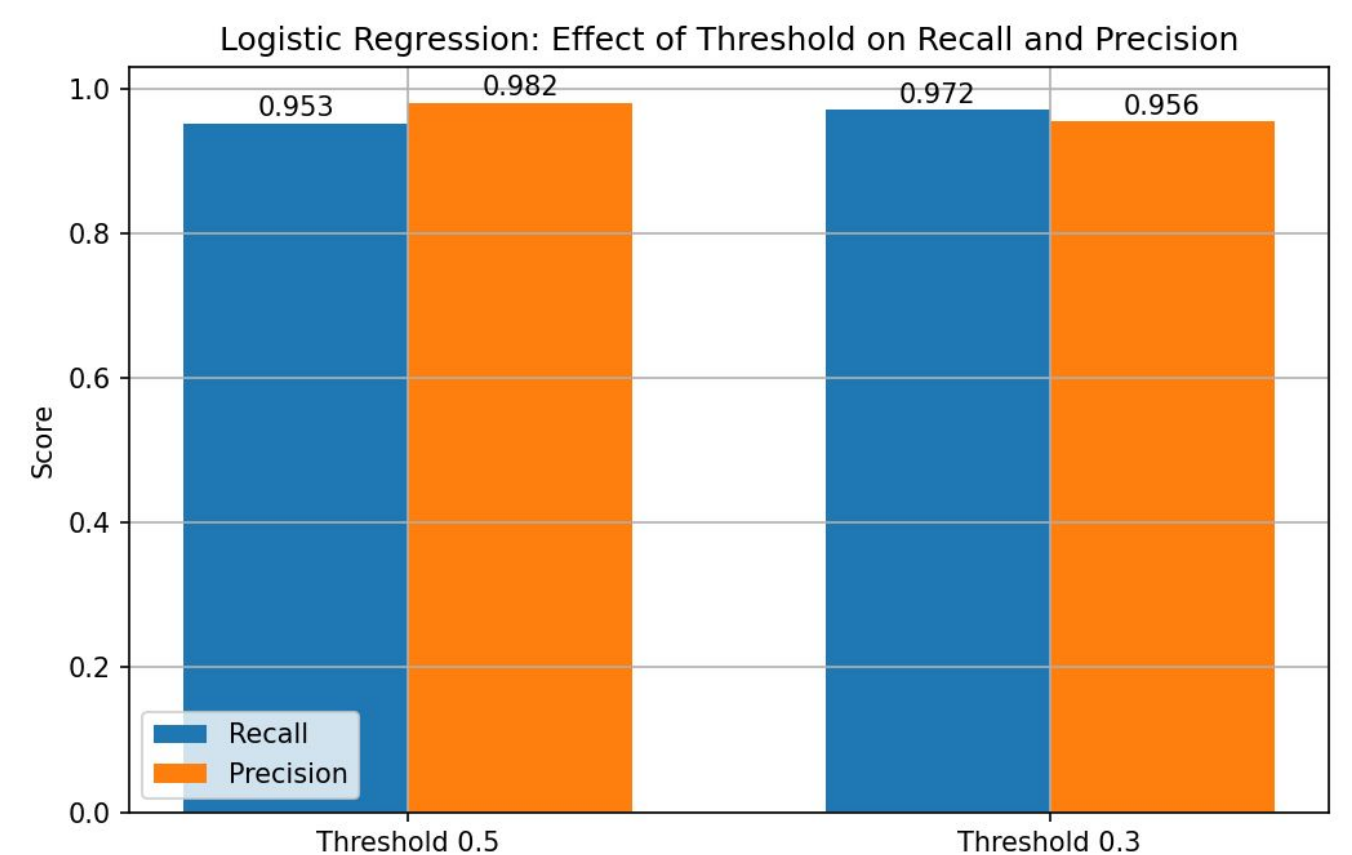


Preprocessing

- Features standardized using training data only
- Prevents data leakage
- Ensurs fair model evaluation

Threshold Analysis (Logistic Regression)

- Lowering threshold (0.5 $\rightarrow$ 0.3)
 - **Recall increases (Sensitivity)**
 - Precision decreases



Hyperparameter Tuning:

